Question			Answer	Marks	AO	Guidance
7	(a)		Total in the AB direction is $3-3=0$ Total in the direction AD is $4+5-9=0$ So zero resultant force	**M1** **A1** **[2]**	**2.1** **2.1**	Considers forces in each direction **AG** Allow if clear statement of zero force in two perpendicular directions, Do not allow for "equilibrium" on its own eg 3=3 and 4+5=9 is not sufficient without a comment
7	(b)		Take moments anticlockwise (N cm) $4\times14-5\times14+3\times9-9\times14+3\times9$ = –86 N cm = –0.86 [N m]	**B1** **M1** **A1** **[3]**	**1.1** **1.1** **1.1**	At least one moment about the centre or any corner Combines moments of 5 forces about the centre. Allow only sign errors Do not allow as final answer Allow for 0.86 N m clockwise but not –0.86 N m clockwise Allow B1 SC1 for -0.86 Nm as the total moment about a different point.
7	(c)		The book will rotate clockwise [about the centre of mass]	**B1** **B1** **[2]**	**2.2a** **1.2**	Allow turn or spin but not move, tilt or flip Allow "there is a clockwise moment" for the second mark